

Lorenz Veithen

Aspiring research engineer devoted to bridging the gap between cutting edge technology and societal challenges.

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TECHNICAL EXPERIENCE

GRADUATE RESEARCHER (MSc THESIS) DELFT UNIVERSITY OF TECHNOLOGY

📍 Delft, The Netherlands 📅 Jan. 2024 - Sep. 2024

- 8-month long research on the tumbling dynamics and attitude control retrieval of solar-sails (awarded 8.5/10).
- Ran large-scale computations on the DelftBlue supercomputer and handled over 400 GB of results data.

AEROTHERMAL ENGINEER REXUS PROGRAMME - TEAM SHEAR (DARE)

📍 Delft, The Netherlands 📅 Jul. 2022 - Jan. 2024

- Flight-proving a new simple to manufacture and cost-efficient heat shield for sounding rocketry through the SHEAR experiment.
- Developing the heat shield production process to improve its manufacturability.

FLIGHT DYNAMICS INTERNSHIP GERMAN AEROSPACE CENTER (DLR)

📍 Munich, Germany 📅 Aug. 2023 - Dec. 2023

- Research project on the Q-Law, a Lyapunov control law to compute near-optimal many-revolutions trajectories between any two bounded orbits.
- Implemented and compared different formulations, devised methods to mitigate thrust chattering, and derived a novel general slot targeting method.
- Presented results at ISSFD2024 conference.

UNDERGRADUATE RESEARCHER DELFT UNIVERSITY OF TECHNOLOGY

📍 Delft, The Netherlands 📅 Sep. 2020 - Aug. 2022

- Developed a novel method to analyse the morphometrics of surface features of meteorites to determine characteristics of its entry.
- Displayed results at the IMC2023 and published as first author in its proceedings.

SPACE SWEEPER PROJECT MANAGER DELFT UNIVERSITY OF TECHNOLOGY

📍 Delft, The Netherlands 📅 April 2022 - Jul. 2022

- Led a team of 10 students in the research and development of a space debris removal mission, leading to the André Kuipers Ruimtevaart prize and a paper presented at the ISSFD2024 conference.
- Worked on systems engineering, orbit design, and telecommunication architecture design.

MISSION DESIGN TEAM LEAD & ENGINEER TEAM TUMBLEWEED

📍 Delft, The Netherlands 📅 Nov. 2019 - May 2022

- Participated in the development of a next-generation Mars rover swarm mission in an international team of 70+ students.
- Defined the mission scientific objectives and architecture, formulated mission and system level requirements, and analysed the Mars Relay Network performance for a wind-driven swarm of rovers.
- From May 2021 onwards, I led a team of 10 engineers towards the first complete Tumbleweed mission feasibility analysis which received a positive review from experts from TU Delft, ESA, and NASA.

EDUCATION

PHD IN PHYSICS UNIVERSITY OF CAMBRIDGE

📍 Cambridge, UK 📅 Oct. 2025 - Present

- Thesis: Unified multiphysics framework for structural, thermal, and electromagnetic load estimations during major disruptions in tokamak nuclear fusion reactors.

MPHIL SCIENTIFIC COMPUTING UNIVERSITY OF CAMBRIDGE

📍 Cambridge, UK 📅 Oct. 2024 - Oct. 2025

GPA: 90.5/100, rank: 1, Distinction

MSc AEROSPACE ENGINEERING DELFT UNIVERSITY OF TECHNOLOGY

📍 Delft, The Netherlands 📅 Sep. 2022 - Sep. 2024

GPA: 8.67/10.0

BSc AEROSPACE ENGINEERING DELFT UNIVERSITY OF TECHNOLOGY

📍 Delft, The Netherlands 📅 Sep. 2019 - Jul. 2022

GPA: 8.64/10.0, Top 5% Cum Laude, Honours student

MINOR FLUIDS & NUMERICAL METHODS ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

📍 Lausanne, Switzerland 📅 Sep. 2021 - Jan. 2022

GPA: 5.6/6.0, Magna Cum Laude

INTERDISCIPLINARY EXPERIENCE

SYNC PROGRAMME PARTICIPANT FOUNDERS AT THE UNIVERSITY OF CAMBRIDGE

📍 Cambridge, UK 📅 Nov. 2024 - May 2025

- Selected as 1 of 30 to build high-impact startups for the benefit of society.

CHALLENGE PROGRAMME PARTICIPANT IDEA LEAGUE

📍 Europe 📅 Sep. 2022 - May 2023

- Selected as 1 of 40 highest achieving students of IDEA League universities.
- Investigated an ill-defined societal problem through political, analytical, economical, and design perspectives.

CERN IDEASQUARE SUMMER SCHOOL CERN

📍 Delft, The Netherlands 📅 May 2021 - Aug. 2021

- Found innovative applications to ATTRACT technologies, built coarse prototypes and pitched the ideas to CERN experts.

SKILLS & INTERESTS

Languages: French: C2 | English: C1 | Dutch: B1

Interests: numerical modelling, space exploration, disruptive concepts, endurance running.

Python MATLAB Fortran C++ AMReX ArcGIS Pro
CATIA Git CFD CUDA LaTeX Office

PUBLICATIONS

-  Veithen, L. and Keller, M. (2024). Predictor-Controller Approach for Q-Law 6th Element Targeting in Low-Thrust Trajectory Design. Proceedings of the 29th International Symposium on Space Flight Dynamics.
-  Bögel E., Buurmeijer H., Veithen L., Meijering F., Alves Teixeira G., Rehling D., Bas Fernández J., van Wolfswinkel P., Zandvliet N., and Struziński J. (2024). Feasibility Analysis of Small-Size Space Debris Removal in Low-Earth Orbit by Space-Based Laser Ablation. Proceedings of the 29th International Symposium on Space Flight Dynamics.
-  Veithen, L. A. V., and de Vet, S. J. (2024). Morphometrics of regmaglypts based on a 3D Model of the fusion-crust of ordinary chondrite Broek in Waterland (L6). In Proceedings of the IMC, Redu, 2023 (pp. 169-176). International Meteor Organization.

AWARDS & HONOURS

- Best performing student of the MPhil in Scientific Computing at the University of Cambridge
- André Kuipers Ruimtevaart Prize 2022.
- Cum Laude (highest obtainable, top 5%) distinction for TU Delft Aerospace Engineering Undergraduate Program.
- Honours student of the TU Delft.
- 1st and 2nd CanSat Belgium competition in 2019/18.